

QSentinel

Quantum-Inspired Cyber Threat Detection for Digital Signature Security

Non-ML Statistical Threat Detection for
Teleportation-Based Quantum Digital Signatures

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Abstract

Quantum Digital Signature (QDS) systems require mechanisms for detecting tampering with quantum communication channels in addition to conventional protocol-level threats such as replay, impersonation, and unauthorized verification. This work presents QSENTINEL, a simulation-based, non-machine-learning threat-detection framework for a teleportation-based QDS setting. The proposed approach embeds a small, secret, session-bound set of Pauli-eigenstate sentinel qubits within the transmitted sequence. Measurements of these sentinels produce an axis-resolved disturbance fingerprint (e_X, e_Y, e_Z) , allowing the system to distinguish directional channel disturbances that may be obscured by a single aggregate error rate.

QSENTINEL first calibrates the expected error level of the honest channel and then applies exact one-sided binomial hypothesis tests independently to the three Pauli axes. The resulting statistical evidence is combined with protocol-level checks for session freshness, replay, verifier authorization, signature consistency, and teleportation consistency. The detector produces three outcomes: ACCEPT, INCONCLUSIVE, and REJECT, with multiple-axis evidence required for rejection.

The framework was evaluated through simulations of X-, Y-, and Z-Pauli attacks, intercept-resend, Z-measure-resend, partial attacks, and channel-noise drift. In the recorded 100-trial experiments, all five principal full-strength attack classes achieved 100% observed detection. A separate 10,000-trial *calibrated-binomial-null* experiment sampled the honest-channel statistical model rather than executing 10,000 full end-to-end quantum simulations, producing 9,728 accepted, 271 inconclusive, and 1 rejected trial. For $n = 333$ sentinels per axis and $\theta_0 = 0.02$, the first one-sided rejection occurs at $k \geq 14$, where $P(K \geq 14) = 0.00799$; $P(K \geq 13) = 0.0180$. That predicts approximately 2.4% INCONCLUSIVE and 1.9×10^{-4} REJECT under the idealized independent three-axis null, compared with the observed 2.71% and 1/10,000.

These results demonstrate the feasibility of using calibrated, axis-resolved statistical evidence for interpretable quantum-channel threat detection under the tested assumptions. QSENTINEL is implemented as an end-to-end software prototype comprising the quantum simulation and detection core, protocol logic, automated tests, FastAPI backend, and web-based dashboard. The work is intended as a practical detection layer and experimental framework rather than a complete production QDS protocol or a formal unconditional or composable security proof.

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1 Problem and Proposed Solution

1.1 Problem

Quantum Digital Signatures require more than ordinary message-integrity checks. In the target teleportation-based setting, the system must detect or flag quantum-channel manipulation while also handling classical protocol threats such as forgery, impersonation, replay and unauthorized verification.

A single error rate is useful for detecting disturbance but loses directional information. Different attacks can produce different patterns across the three Pauli axes. The problem addressed by QSENTINEL is therefore:

How can a software system obtain interpretable quantum evidence of channel tampering, compare it against an honest noise floor, and combine that evidence with protocol-level security checks without using machine learning?

The quantum foundations follow the teleportation protocol of Bennett *et al.* [3], while the broader QDS setting is grounded in the foundational Gottesman–Chuang construction [1] and later QKD-component QDS work [2].

1.2 Solution at a Glance

The proposed workflow is:

1. generate a signature/session and derive a secret, session-bound sentinel schedule;
2. transmit ordinary payload states together with hidden X/Y/Z Pauli-eigenstate sentinels;
3. model teleportation, channel noise and attack behaviour;
4. measure sentinels and estimate (e_X, e_Y, e_Z) ;
5. compare each axis against its calibrated honest baseline using an exact one-sided binomial test;
6. combine the quantum result with replay, authorization, freshness and signature checks;
7. expose the result through a web dashboard and log security events.

1.3 Why the Approach is Non-ML

The detector does not train a classifier. Its evidence comes directly from measured sentinel outcomes, a calibrated null model, and an explicit statistical decision rule. This makes the reason for a decision inspectable: the system can show which axis exceeded the expected honest-channel error level and why the result became inconclusive or reject.

2 Quantum and Protocol Foundations

2.1 Pauli Eigenstates and the Fingerprint

The Pauli operators are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Their eigenstates define the X, Y and Z measurement axes. A Pauli operation leaves its own eigenstates unchanged up to phase while disturbing eigenstates of the other axes. Consequently, strong Pauli attacks produce characteristic fingerprints.

Table 1: Idealized error fingerprints without channel noise.

Condition	e_X	e_Y	e_Z
None	0	0	0
X-Pauli	0	1	1
Y-Pauli	1	0	1
Z-Pauli	1	1	0
Intercept-resend	$\approx 1/3$	$\approx 1/3$	$\approx 1/3$
Z-measure-resend	$\approx 1/2$	$\approx 1/2$	≈ 0

These are ideal fingerprints, not guaranteed experimental values. Honest channel noise shifts the measured rates upward, which is why QSENTINEL calibrates a baseline before detection. Test-state and authentication ideas are consistent with the broader literature on quantum-message authentication [4].

2.2 Teleportation Model

The simulator follows the standard teleportation structure: an input state is combined with an entangled Bell pair, a Bell measurement yields two classical bits, and the receiver applies the corresponding Pauli correction [3]. In compact form, the pre-correction receiver state is represented by

$$X^{m_2} Z^{m_1} |\psi\rangle, \quad (m_1, m_2) \in \{0, 1\}^2.$$

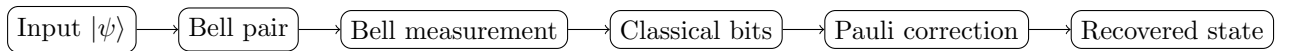


Figure 1: Simplified teleportation structure represented in the simulator.

2.3 Protocol Security Controls

The prototype separates quantum evidence from protocol evidence:

- **Replay:** a previously used session/signature is rejected rather than treated as a new transmission.
- **Verifier authorization:** the verifier identity must match the registered verifier for the session.
- **Freshness:** the sentinel schedule is derived from secret key material and the session identifier using HMAC-SHA256.
- **Wrong-key verification:** verification with incorrect key material produces strong sentinel disturbance in the tested setting.

- **Teleportation consistency:** Bell outcomes and Pauli corrections are part of the simulated workflow.

3 QSentinel Detection Method

3.1 Secret Session-Bound Sentinels

For each session, QSENTINEL derives the sentinel positions from secret key material and a session identifier. The derivation uses HMAC-SHA256 with the context string `QSentinel-Sentinel-Schedule-v2:`. Thus, the schedule is reproducible for the authorized session but changes when the session changes.

A recorded freshness check gave

$$A = B : \text{False}, \quad A = C : \text{True},$$

where A and B use different session identifiers and C repeats A 's session.

3.2 Measurement and Error Rates

Let n_a be the number of sentinels measured on axis $a \in \{X, Y, Z\}$ and k_a the number of errors. The empirical error rate is

$$\hat{e}_a = \frac{k_a}{n_a}.$$

The resulting vector $(\hat{e}_X, \hat{e}_Y, \hat{e}_Z)$ is the quantum fingerprint used by the detector.

For an honest depolarizing channel with parameter p , the simulated per-axis error probability is

$$\theta = \frac{2p}{3}.$$

The system estimates the honest baseline through calibration sessions rather than assuming the theoretical value is exact. In the controlled experiments reported here, the calibrated baseline was fixed at 0.02 per axis so that detector behavior could be compared under repeatable conditions. Calibration uncertainty is not propagated into the current exact binomial test; handling that uncertainty explicitly is left as future work.

3.3 Statistical Decision Rule

For each axis, the detector tests

$$H_0 : e_a \leq b_a \quad \text{vs.} \quad H_1 : e_a > b_a,$$

where b_a is the calibrated honest baseline. An exact one-sided binomial p -value is compared with significance level α .

Wilson intervals are used for descriptive uncertainty reporting [13], while Clopper–Pearson intervals are used when reporting exact confidence bounds for binomial proportions [14].

The final detector is intentionally conservative:

- 0 rejected axes $\rightarrow$ ACCEPT;
- 1 rejected axis $\rightarrow$ INCONCLUSIVE;
- 2 or more rejected axes $\rightarrow$ REJECT.

This design avoids treating a single noisy axis as sufficient evidence of an attack.

3.4 Meaning of INCONCLUSIVE

INCONCLUSIVE is a deliberate third outcome, not a failure of the detector. It means the observed evidence is abnormal enough to require caution but does not satisfy the multi-axis rejection rule. This is particularly relevant for weak attacks and channel drift.

4 Attack Models

The simulator implements both directional and basis-independent disturbance models.

Table 2: Attack models used in evaluation.

Attack	Modelled behaviour
X/Y/Z-Pauli	Applies the corresponding Pauli operation to targeted states.
Intercept-resend	Intercepts and resends according to a measurement/resend model, producing approximately balanced disturbance across axes.
Z-measure-resend	Measures in the Z basis and resends the resulting state; X/Y sentinels are strongly disturbed while Z is comparatively preserved.
Partial attack	Applies an attack with configurable strength to study statistical detection power.
Depolarizing noise	Represents honest-channel noise and controlled noise drift.

The attack models are deliberately explicit and interpretable. They are not intended to represent every possible adaptive quantum adversary.

5 Implementation and Public Prototype

5.1 Software Architecture

The implementation is organized into separable modules for quantum states, attacks, sentinel scheduling and measurement, noise, detection, teleportation, protocol logic and the QDS workflow. Experiments are kept separately so that calibration, robustness and statistical validation can be reproduced independently.

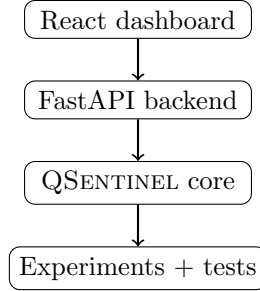


Figure 2: Software architecture of the prototype.

5.2 Core Repository Structure

```

QSentinel/
src/ states, attacks, sentinels, detector,
      teleportation, protocol, QDS pipeline
experiments/ calibration, attack, robustness and evaluation scripts
backend/ FastAPI application and deployment requirements
QSentinel_Frontend_Animated/
      React dashboard
tests/ automated tests
  
```

5.3 Backend and Dashboard

The backend exposes signature creation and verification, replay simulation, attack simulation, forgery/impersonation scenarios, unauthorized-verification simulation, and security-event logging. The public dashboard provides signature creation, clean verification, replay testing and a Threat Lab for attack scenarios.

Table 3: Main public API operations.

Route	Purpose
/signature/create	Create a simulated signature/session
/signature/verify	Verify a signature
/signature/replay	Test replay handling
/simulate	Run an attack/noise scenario
/simulate/replay	Simulate replay
/simulate/unauthorized	Simulate unauthorized verification
/simulate/forgery	Wrong-key forgery simulation
/simulate/impersonation	Impersonation scenario
/events	Security-event logging

5.4 Deployment

The public prototype uses a GitHub Pages React frontend and a Render-hosted FastAPI backend. The backend is a free-tier deployment and may require a short wake-up period after inactivity.

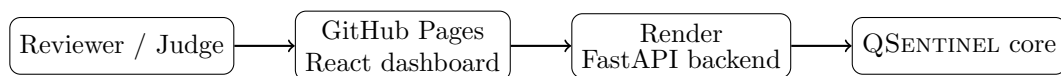


Figure 3: Public deployment architecture.

Public resources:

- <https://dwibon.github.io/QSentinel/>
- <https://dwibon.github.io/QSentinel/verify>
- <https://github.com/Dwibon/QSentinel>

For live demonstrations, the frontend should request the backend health endpoint (`/health`) when the page loads. While the Render service is waking, the interface should display a short “waking server...” status rather than appearing unresponsive. The backend should also be manually warmed before a recorded demonstration. The present deployment uses Render’s free tier, so a first request after inactivity can take roughly 50–60 seconds.

6 Experimental Evaluation

6.1 Evaluation Configuration

Unless otherwise stated, the main quantitative experiments used $n = 5000$ qubits, a sentinel fraction of 20%, honest-channel noise $p = 0.03$, baseline 0.02 per axis, and $\alpha = 0.01$. Detection-rate experiments used 100 trials per condition.

With 20% of 5000 qubits used as sentinels, approximately 1000 sentinel measurements are available per trial, or approximately $n = 333$ per axis. This value is used for the calibrated-binomial-null calculation below.

6.2 Representative Fingerprints

A representative integrated run produced:

Table 4: Representative end-to-end fingerprints.

Condition	e_X	e_Y	e_Z	Decision
None	.020	.028	.021	ACCEPT
X-Pauli	.009	.981	.988	REJECT
Y-Pauli	.976	.021	.962	REJECT
Z-Pauli	.979	.968	.006	REJECT
Intercept-resend	.344	.301	.366	REJECT
Z-measure-resend	.522	.500	.022	REJECT

6.3 Full-Strength Attack Detection

For X-Pauli, Y-Pauli, Z-Pauli, intercept-resend and Z-measure-resend, each condition achieved 100/100 observed detection in the recorded 100-trial experiments. For a 100/100 result, the two-sided 95% Clopper–Pearson lower bound is approximately 96.38%; the result is therefore empirical rather than a universal guarantee [14].

6.4 Honest-Channel Validation

A separate 10,000-trial *calibrated-binomial-null experiment* sampled the calibrated honest-channel model; it did not execute 10,000 full end-to-end quantum simulations. The observed outcomes were:

Table 5: 10,000-trial calibrated-binomial-null validation.

Outcome	Count	Observed rate
ACCEPT	9,728	97.28%
INCONCLUSIVE	271	2.71%
REJECT	1	0.01%

For $n = 333$ and $\theta_0 = 0.02$, the first one-sided rejection occurs at $k \geq 14$:

$$P(K \geq 14) = 0.00799, \quad P(K \geq 13) = 0.0180.$$

Thus an individual axis is rejected only when at least 14 errors are observed. With three independent axes under the idealized calibrated null, the probability of exactly one rejected axis is approximately 2.4%, while the probability of two or more rejected axes is approximately

1.9×10^{-4} . These values predict roughly 2.4% INCONCLUSIVE and 2×10^{-4} REJECT under the simplified null model. The observed 2.71% and 1/10,000 rates are close in scale but are not a replacement for the theoretical calculation.

The observed rejection rate was $1/10,000 = 0.01\%$. This is an observed null-sampling rate, not a theoretical false-positive guarantee.

6.5 Wrong-Key Verification

In 100 wrong-key trials, the mean fingerprint was

$$(0.4985, 0.5069, 0.5020),$$

with 100 REJECT, 0 INCONCLUSIVE and 0 ACCEPT. This experiment demonstrates wrong-key verification behaviour; it should not be interpreted as a complete proof of signer impersonation resistance.

6.6 Partial-Attack Power

Table 6: Observed detection power for partial attacks; 100 trials per condition.

Strength	X	Y	Z	Intercept	Z-measure
0.01	.020	.000	.000	.000	.000
0.02	.190	.140	.190	.000	.000
0.05	.910	.900	.930	.150	.430
0.10	1.000	1.000	1.000	.930	.950
0.20+	1.000	1.000	1.000	1.000	1.000

The results illustrate the statistical nature of the system: weak attacks can remain below the rejection threshold, while stronger attacks become increasingly detectable.

6.7 Noise Drift

With a calibrated baseline of 0.02, 100 honest trials were run at each noise level. For the depolarizing model, the corresponding theoretical per-axis value is $\theta = 2p/3$.

Table 7: Observed decisions under honest-channel noise drift.

p	$\theta = 2p/3$	ACCEPT	INCONCLUSIVE	REJECT
0.020	.0133	100	0	0
0.022	.0147	100	0	0
0.025	.0167	99	1	0
0.030	.0200	94	6	0
0.035	.0233	95	5	0
0.040	.0267	79	20	1

Across the tested drift points, the dominant observed effect was movement toward INCONCLUSIVE. At $p = 0.04$, one of the 100 honest trials was REJECT. This single observed rejection should not be interpreted as evidence of a general drift threshold; it motivates periodic recalibration and larger statistical studies.

6.8 Fingerprint Visualization

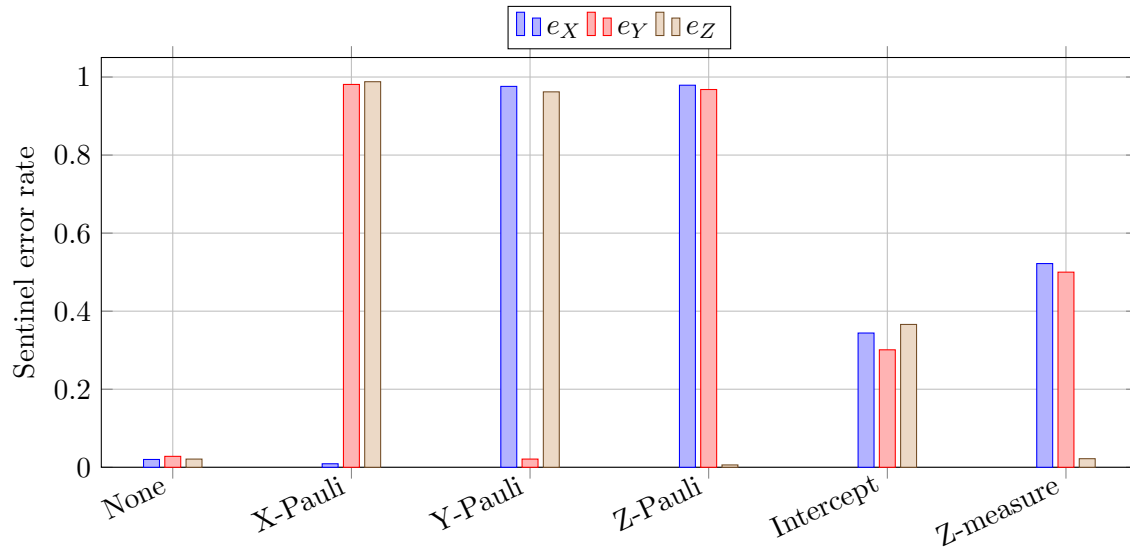


Figure 4: Representative axis-resolved fingerprints used by the detector.

7 Security Interpretation and Limits

7.1 Why Axis-Resolved Evidence Helps

The vector (e_X, e_Y, e_Z) preserves directional information that an aggregate error rate discards. A strong X-Pauli attack, for example, leaves the X eigenstates comparatively stable while strongly disturbing Y and Z sentinels. This makes the evidence interpretable rather than merely anomalous.

7.2 Closest Related Work and Positioning

QDS constructions include the foundational Gottesman–Chuang model [1] and QDS protocols using QKD components [2]. Quantum-assisted signature work such as Tarable *et al.* instead derives information-theoretic security from QKD keys and uses Carter–Wegman MACs; it is therefore a QKD-key hybrid rather than a teleportation-based detection scheme [6].

Several related schemes are specifically arbitrated quantum-signature (AQS) protocols, including controlled-teleportation designs [7], security analyses of AQS schemes [9], and recent cryptanalysis of multiple AQS constructions [10]. The latter work includes the controlled-teleportation scheme of Lu *et al.* among the schemes it analyzes [10].

The distinction is important: QSENTINEL is not proposed as another arbitrated signature construction. Its intended contribution is a detection layer for a teleportation-based QDS setting. Its methodological distinction is the use of secret, session-bound Pauli-eigenstate sentinels to obtain an axis-resolved disturbance vector, rather than relying only on a single aggregate error rate.

The controlled-teleportation literature also uses mutually unbiased-basis decoy particles for eavesdropping detection and identity authentication [7]. QSENTINEL instead emphasizes statistical use of the three-axis aggregate fingerprint (e_X, e_Y, e_Z) and a conservative multi-axis decision rule.

A further QDS construction uses superdense teleportation and does not require an arbitrator [12]. It is therefore another example of why QDS and AQS should not be treated as interchangeable protocol models.

7.3 Identifiability Limit

For a general channel represented by a Bloch-vector map T , the error observed by a Pauli-axis eigenstate satisfies

$$e_a = \frac{1 - T_{aa}}{2},$$

when the $+$ and $-$ eigenstates of the corresponding Pauli axis are sampled with balanced signs. Therefore, two different channels with the same diagonal elements can produce the same three sentinel error rates. The fingerprint is consequently a useful detector feature, not a complete characterization of an arbitrary quantum channel.

7.4 Security Assumptions

The prototype assumes that the secret key material and session binding used for sentinel scheduling are protected, the authorized verifier can identify the expected sentinel bases, and the statistical null model reasonably represents honest operation. It also assumes the tested attack/channel models are relevant to the scenario being evaluated.

7.5 What QSentinel Does Not Claim

QSENTINEL does not provide a universal detector for all quantum attacks, a composable security proof, a complete production QDS/QKD protocol, or a hardware-level security guarantee.

Replay and verifier authorization are protocol controls rather than direct quantum-channel measurements. Experimental detection rates are observed simulation results.

7.6 Holevo Information: Correct Scope

Holevo information can be used to quantify an upper bound on accessible classical information about a quantum ensemble under stated assumptions. It is not itself a detector and does not by itself provide a forging probability. Any stronger claim about guessing probability requires an explicit min-entropy or equivalent security analysis.

8 Testing, Reproducibility and Limitations

8.1 Automated Testing

The recorded test suite contains 78 passing tests. It covers state preparation and measurement, attacks, noise, sentinel scheduling, detection decisions, teleportation/protocol logic and selected API-facing behaviour. The freshness test confirms that different session identifiers produce different schedules while the same session reproduces the schedule.

The recorded count should not be interpreted as complete endpoint coverage: the current report does not claim that the newly exposed unauthorized-verification, forgery and impersonation paths are each exhaustively covered by dedicated automated tests.

8.2 Reproduce the Prototype

```
git clone https://github.com/Dwibon/QSentinel.git
cd QSentinel
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
python -m pytest tests/

# backend
uvicorn backend.app:app --reload

# frontend, in another terminal
cd QSentinel_Frontend_Animated
npm install
npm run dev
```

Local backend: <http://127.0.0.1:8000>.

Local frontend: <http://localhost:5173>.

8.3 Current Limitations

- The quantum processor is simulated; results are not hardware measurements.
- The signature construction is a prototype abstraction rather than a complete production QDS construction.
- The integer key represents secret material; a complete BB84/QKD protocol is not implemented.
- Detection depends on calibration and can become inconclusive or inaccurate under substantial noise drift.
- The evaluated attack set is finite; adaptive or correlated attacks may behave differently.
- The current detector does not establish a formal security theorem.
- Calibration uncertainty is not yet propagated through the exact binomial decision rule.

9 Current Contribution and Future Work

9.1 Contribution Positioning

QSENTINEL builds on established concepts from quantum teleportation, quantum digital signatures, quantum authentication/test-state methods, Pauli operators and statistical inference. The project does not claim these individual components as novel.

The integrated contribution is a system-level methodology that combines:

1. secret session-bound sentinel scheduling;
2. three-axis quantum disturbance fingerprints;
3. honest-channel calibration;
4. exact per-axis statistical testing with a conservative multi-axis rule;
5. explicit simulation of several attack classes and partial attack strength;
6. protocol-level replay and authorization checks;
7. an end-to-end API and dashboard suitable for reproducible demonstration.

The contribution is therefore positioned as a detection and evaluation framework rather than a replacement for the underlying QDS protocol.

9.2 Future Work

- Validation on actual quantum hardware and broader quantum-channel models.
- Evaluation against more sophisticated correlated, adaptive, and coherent attacks.
- Formal security analysis of detection guarantees and error thresholds.
- Adaptive calibration to handle changing noise levels and channel drift.
- Larger-scale statistical evaluation across more qubits and repeated trials.
- Integration with a complete QDS protocol implementation.
- Hardware-backed verifier authentication.
- Extension to additional quantum observables beyond the three Pauli axes.

10 Conclusion

QSENTINEL addresses the practical detection problem in the SIH setting by turning hidden Pauli-eigenstate test transmissions into an interpretable three-axis statistical fingerprint. Calibration makes the detector sensitive to the honest noise floor, while the INCONCLUSIVE state avoids forcing every borderline observation into accept or reject. Protocol checks cover threats that quantum-channel statistics alone cannot establish, such as replay and verifier authorization.

The current prototype demonstrates the complete path from signature/session generation through teleportation and attack simulation to statistical detection, API exposure, dashboard visualization and logging. Its experimental results support the feasibility of the proposed methodology under the tested assumptions while also defining clear limits on what can be claimed. The principal next steps are hardware validation, broader attack coverage, explicit treatment of calibration uncertainty, and formal security analysis.

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